

SIH26153

PRD • TRD • Architecture • MVP Core Features

Product concept: A passive, temporal cyber-threat forecasting platform that learns from ordered network-flow history and predicts likely next-stage attacker behavior without actively probing monitored systems.

1. PRD — Product Requirements

Problem

Conventional IDS workflows are mainly detection-oriented: they identify suspicious activity after recognizable behavior appears. This prototype adds a forecasting layer that uses recent temporal context to estimate what an attacker is likely to do next.

Goal

Build a demonstrable MVP that ingests network-flow data, maintains a per-host temporal state, forecasts the next likely security stage, explains the forecast with measurable evidence, and replays a held-out attack sequence so judges can observe the warning before the labeled event.

Target Users

User	Primary need
SOC / security analyst	See current trajectory and likely next stage.
Incident responder	Get early warning and supporting evidence.
Security administrator	Replay incidents and inspect model performance.

Functional Requirements

ID	Requirement	Priority
FR-01	Ingest and normalize CICFlowMeter/CIC-IDS flow records.	Must
FR-02	Clean null, infinite, duplicate and malformed records.	Must
FR-03	Aggregate flows into fixed windows per host.	Must
FR-04	Create chronological, leakage-safe sequences.	Must
FR-05	Provide a single-window baseline classifier.	Must
FR-06	Forecast future state / next stage from temporal history.	Must
FR-07	Map forecast to an interpretable attack stage.	Must
FR-08	Show confidence, lead time and evidence.	Must
FR-09	Replay a held-out attack sequence.	Must
FR-10	Perform out-of-distribution evaluation.	Should

2. TRD — Technical Requirements

Dataset Strategy

Primary: CSE-CIC-IDS2018 and its CICFlowMeter-derived flow records.

Temporal variant: The proposed IDS2018-APT hand-injected multi-stage sequence data, where available, to provide explicit campaign ordering such as reconnaissance → lateral movement → exfiltration.

OOD test: CIC-IDS2017 reserved for final generalization testing and not used for training.

Split: Chronological split is preferred. The proposed days 1–7 train / days 8–10 test plan must be applied only after confirming the actual files contain those day partitions and labels.

ML / Analytics

Baseline: Logistic Regression or Random Forest using one window, providing the non-temporal comparison.

Temporal model: Start with a small LSTM. It consumes N previous host-level state vectors and predicts a future state / next stage. A Transformer is an optional experiment, not an MVP dependency.

Stage mapping: Deterministic interpretation layer using measurable network-flow indicators to map predictions to stages such as reconnaissance, lateral movement, command-and-control, and exfiltration.

Evidence: SHAP or another valid attribution method to identify influential input features. The UI must clearly separate observed evidence from inferred forecast.

Recommended Stack

Layer	Technology	Purpose
Data	Python, pandas, NumPy	Parsing, cleaning, windowing, sequences
Baseline ML	scikit-learn	Classifier and evaluation
Temporal ML	PyTorch	LSTM forecasting
Explainability	SHAP / attribution	Feature-level evidence
Backend	FastAPI (optional)	Model/replay API
Dashboard	React or Streamlit	Analyst UI and replay
Charts	Plotly / native charts	Trajectory and evidence visuals

Evaluation

Classification: precision, recall, F1, false-positive rate and confusion matrix. Forecasting: next-stage accuracy, warning lead time, forecast precision and false forecast rate. Generalization: performance delta between the normal test set and CIC-IDS2017.

3. Architecture

High-Level Pipeline

Raw flow records → ingestion/validation → cleaning & normalization → fixed time windows → host temporal sequences → baseline + temporal model → stage mapping → confidence/evidence → dashboard → replay/evaluation.

Components

Component	Responsibility
Ingestion	Load flow CSVs, validate schema, normalize fields.
Preprocessor	Clean data, scale/select features and construct windows.
Sequence builder	Build chronological host sequences without future leakage.
Baseline engine	Classify the current window without temporal context.
Forecast engine	Use recent windows to predict future state/stage.
Stage mapper	Translate output into interpretable security stages.
Evidence engine	Extract influential features and measurable signals.
Replay engine	Stream held-out sequences and reveal ground truth after forecast.
Dashboard	Display state, forecast, confidence, lead time, trajectory and evidence.

Data Flow

1. Load flow record → 2. validate → 3. clean/normalize → 4. aggregate into host window → 5. append to temporal buffer → 6. baseline prediction → 7. LSTM forecast → 8. stage interpretation → 9. evidence extraction → 10. dashboard update → 11. ground-truth reveal and metric calculation during replay.

Architecture Constraints

The MVP is passive: it does not scan, probe, exploit or modify monitored hosts. Model outputs are predictions, not confirmed incidents. Demo values must come from the actual replay/evaluation pipeline.

4. MVP Core Features

#	Feature	MVP outcome
1	Passive flow ingestion	Real flow records become ordered security windows.
2	Temporal host timeline	Shows how observed behavior changes over time.
3	Current-stage detection	Shows what the system detects now.
4	Next-stage forecasting	Predicts likely next attack stage before the event.
5	Confidence + lead time	Reports model confidence and measured warning interval.
6	Evidence / Why panel	Shows features/signals contributing to the forecast.
7	Baseline comparison	Demonstrates value of temporal context over single-window detection.
8	Attack replay	Forecast first, then reveal the labeled event.
9	Performance panel	Shows actual evaluation metrics.
10	OOD validation	Tests generalization on CIC-IDS2017, if feasible within MVP time.

5. Demo Scenario

Choose a genuinely held-out multi-stage sequence. Start the replay before the target transition. The system displays observed history, then forecasts the next stage with a confidence value. Continue the replay until the actual labeled event appears. Record the warning interval. Show the non-temporal baseline beside it to demonstrate why temporal context matters.

Non-negotiable: Do not fabricate confidence, lead time, accuracy, or “forecast-before-event” claims. Every number shown in the pitch should be reproducible from the actual evaluation pipeline.

6. Scope Control

Build first: data pipeline, chronological split, baseline, LSTM, stage mapping, evidence, replay and dashboard.

Defer: large Transformers, live enterprise packet capture, active response, unnecessary cloud infrastructure, complex multi-agent AI and hardware dependencies.

7. Definition of Done

The MVP is complete when preprocessing is reproducible; the split is leakage-safe; baseline and temporal models train; a held-out sequence generates a forecast before its labeled future event; lead time is measured automatically; evidence is tied to model inputs; the dashboard uses actual outputs; and weaknesses are reported honestly.